

REPORT

ON

**SOC Lab Environment with Wazuh, Web Application
Firewall(WAF), and DVWA**

1. Summary

This project successfully designed and implemented a Security Operations Center (SOC) lab environment using VirtualBox. The setup consisted of an Ubuntu server running Apache, ModSecurity with OWASP Core Rule Set (CRS), and DVWA as a vulnerable web application. A Wazuh stack (Manager, Indexer, Dashboard) was deployed for log collection, correlation, and visualization. A Kali Linux machine was used as the attacker system to simulate real-world threats such as SQL Injection, Cross-Site Scripting (XSS), and brute force attacks.

Testing showed that ModSecurity was able to detect and block malicious web requests at the application layer, while Wazuh ingested and correlated logs for centralized monitoring. The lab demonstrated how a WAF and SIEM solution can work together in a SOC environment to enhance visibility, detection, and response.

2. Project Objectives and Purpose

- Build and configure a SOC on Ubuntu.
 - Deploy a Web Application Firewall (WAF) using ModSecurity and OWASP CRS.
 - Install and configure Wazuh for centralized security monitoring and alerting.
 - Host vulnerable applications (DVWA).
 - Launch known attacks (SQL Injection, XSS, brute force) from Kali Linux.
 - Observe and analyze alerts generated and blocked requests in Wazuh Dashboard and WAF logs.
 - Enhance practical SOC, SIEM, and blue team skills.
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3. Environment Setup & Methodology

3.1 Lab Components

- **Ubuntu GUI 22.04** (Main SOC node, hosting Apache, WAF, and DVWA)
- **Kali Linux VM** (Attacker, for simulating exploits)
- **Wazuh Manager, Indexer, Dashboard** (Security monitoring and log analysis)
- **DVWA** (Target vulnerable web app)
- **ModSecurity + OWASP CRS** (Web Application Firewall)

3.2 Key Installation Commands

Wazuh Installer

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
```

Apache + WAF (ModSecurity + CRS)

```
sudo apt update
```

```
sudo apt install apache2
```

```
sudo apt install libapache2-mod-security2
```

```
sudo mv /etc/modsecurity/modsecurity.conf-recommended  
/etc/modsecurity/modsecurity.conf
```

```
sudo nano /etc/modsecurity/modsecurity.conf
```

(Edit the Conf file and Set the **SecRuleEngine** to **ON**)

```
sudo a2enmod security2
```

```
sudo systemctl restart apache2
```

```
sudo apt install modsecurity-crs
```

```
sudo ln -s /usr/share/modsecurity-crs /usr/share/modsecurity-crs/activated_rules
```

```
sudo apachectl configtest
```

DVWA

DVWA install

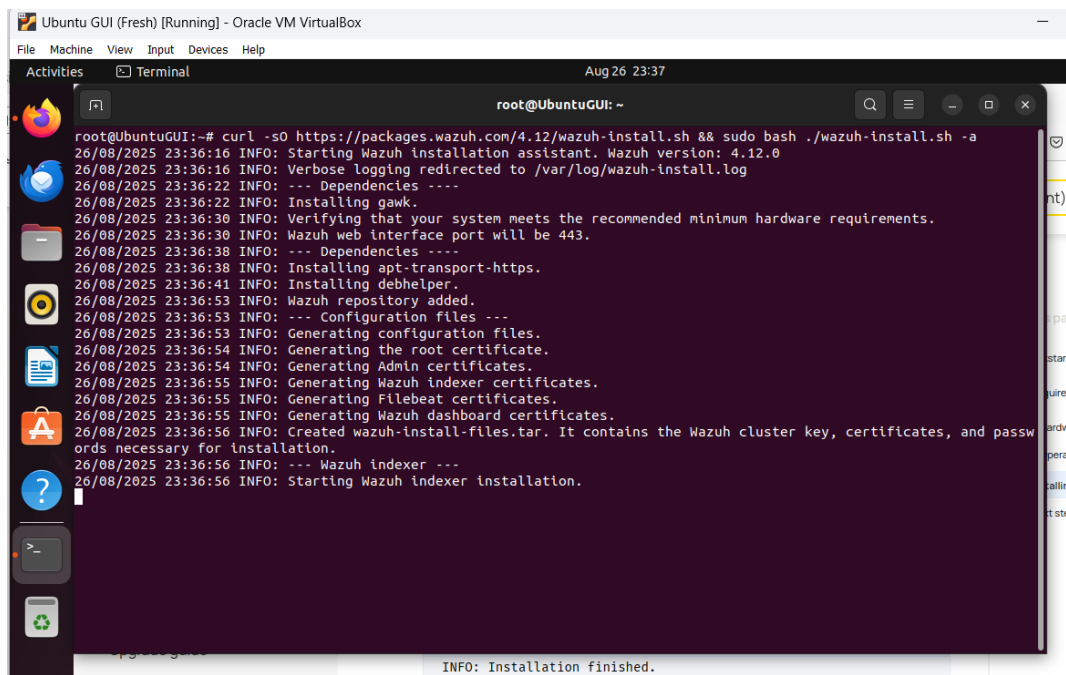
```
sudo bash -c "$(curl --fail --show-error --silent --location  
https://raw.githubusercontent.com/IamCarron/DVWA-Script/main/Install-  
DVWA.sh)"
```

3.4 Step by step instruction to install Wazuh Dashboard and Agent

A. Installing Wazuh (Manager, Indexer, Dashboard) on Ubuntu GUI

1. Download and Run Wazuh Installer Script

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-  
install.sh -a
```



The screenshot shows a terminal window titled "Ubuntu GUI (Fresh) [Running] - Oracle VM VirtualBox". The terminal output displays the execution of the Wazuh installer script. The process begins with downloading the script and then proceeds through several steps: starting the installation assistant, logging redirection, dependency checks, installing gawk, verifying hardware requirements, adding the Wazuh repository, and generating configuration files and certificates. The final output indicates that the installation is finished.

```
root@UbuntuGUI:~# curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
26/08/2025 23:36:16 INFO: Starting Wazuh installation assistant. Wazuh version: 4.12.0
26/08/2025 23:36:16 INFO: Verbose logging redirected to /var/log/wazuh-install.log
26/08/2025 23:36:22 INFO: --- Dependencies ---
26/08/2025 23:36:22 INFO: Installing gawk.
26/08/2025 23:36:30 INFO: Verifying that your system meets the recommended minimum hardware requirements.
26/08/2025 23:36:30 INFO: Wazuh web interface port will be 443.
26/08/2025 23:36:38 INFO: --- Dependencies ---
26/08/2025 23:36:38 INFO: Installing apt-transport-https.
26/08/2025 23:36:41 INFO: Installing debhelper.
26/08/2025 23:36:53 INFO: Wazuh repository added.
26/08/2025 23:36:53 INFO: --- Configuration files ---
26/08/2025 23:36:53 INFO: Generating configuration files.
26/08/2025 23:36:54 INFO: Generating the root certificate.
26/08/2025 23:36:54 INFO: Generating Admin certificates.
26/08/2025 23:36:55 INFO: Generating Wazuh indexer certificates.
26/08/2025 23:36:55 INFO: Generating Filebeat certificates.
26/08/2025 23:36:55 INFO: Generating Wazuh dashboard certificates.
26/08/2025 23:36:56 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passw
ords necessary for installation.
26/08/2025 23:36:56 INFO: --- Wazuh indexer ---
26/08/2025 23:36:56 INFO: Starting Wazuh indexer installation.
INFO: Installation finished.
```

2. Note down the admin password for first login

```
26/08/2025 23:39:18 INFO: Wazuh indexer cluster security configuration initialized.
26/08/2025 23:39:18 INFO: Wazuh indexer cluster initialized.
26/08/2025 23:39:18 INFO: --- Wazuh server ---
26/08/2025 23:39:18 INFO: Starting the Wazuh manager installation.
26/08/2025 23:42:42 INFO: Wazuh manager installation finished.
26/08/2025 23:42:42 INFO: Wazuh manager vulnerability detection configuration finished.
26/08/2025 23:42:42 INFO: Starting service wazuh-manager.
26/08/2025 23:43:00 INFO: wazuh-manager service started.
26/08/2025 23:43:00 INFO: Starting Filebeat installation.
26/08/2025 23:43:19 INFO: Filebeat installation finished.
26/08/2025 23:43:21 INFO: Filebeat post-install configuration finished.
26/08/2025 23:43:21 INFO: Starting service filebeat.
26/08/2025 23:43:22 INFO: filebeat service started.
26/08/2025 23:43:22 INFO: --- Wazuh dashboard ---
26/08/2025 23:43:22 INFO: Starting Wazuh dashboard installation.
26/08/2025 23:48:06 INFO: Wazuh dashboard installation finished.
26/08/2025 23:48:06 INFO: Wazuh dashboard post-install configuration finished.
26/08/2025 23:48:06 INFO: Starting service wazuh-dashboard.
26/08/2025 23:48:06 INFO: wazuh-dashboard service started.
26/08/2025 23:48:09 INFO: Updating the internal users.
26/08/2025 23:48:18 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
26/08/2025 23:48:38 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
26/08/2025 23:49:14 INFO: Initializing Wazuh dashboard web application.
26/08/2025 23:49:15 INFO: Wazuh dashboard web application initialized.
26/08/2025 23:49:15 INFO: --- Summary ---
26/08/2025 23:49:15 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
  User: admin
  Password: 508-00400W-1-17X77C31-014-N361
26/08/2025 23:49:15 INFO: --- Dependencies ---
26/08/2025 23:49:15 INFO: Removing gawk.
26/08/2025 23:49:20 INFO: Installation finished.
```

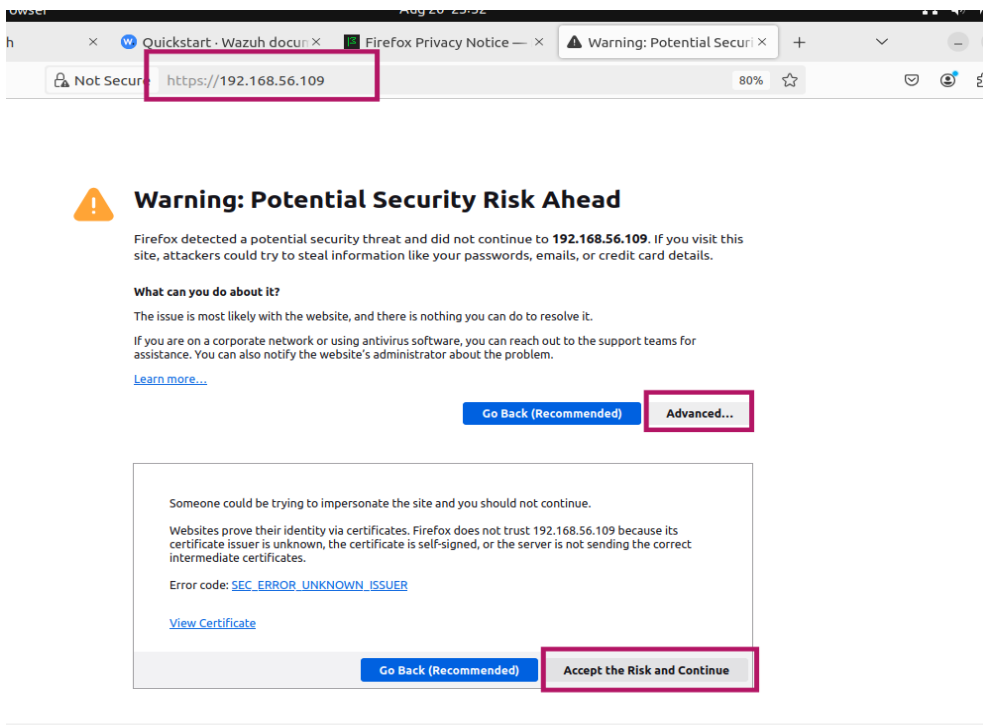
Note: If you forgot to note down the password it can be seen by running this command:

```
sudo tar -O -xvf wazuh-install-files.tar wazuh-install-files/wazuh-  
passwords.txt
```

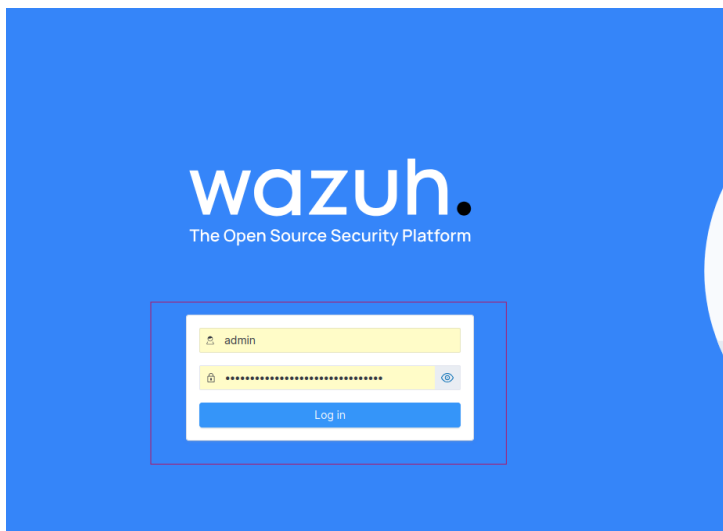
3. Access Wazuh Dashboard

- Open browser and go to
 - <http://localhost> OR https://<your_server_ip>

- After going to the website Click on Advanced and Select Accept Risk and

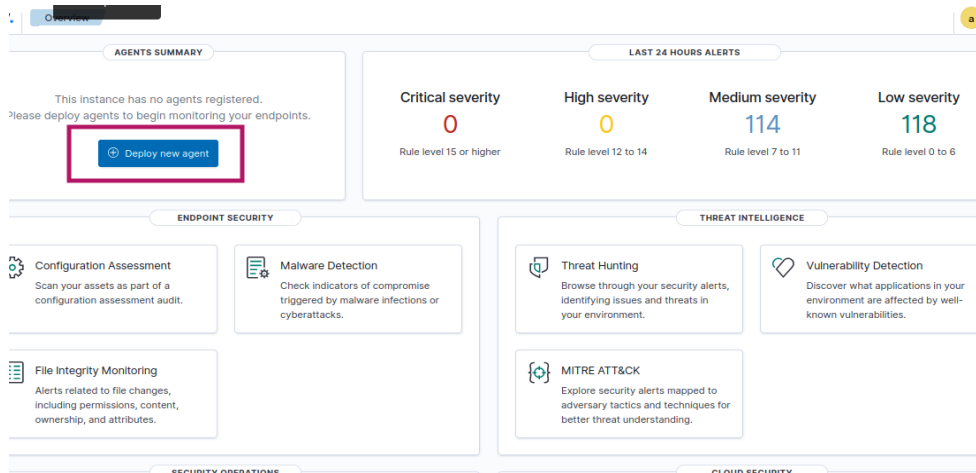


- Login to Wazuh with username as admin and password copied from above

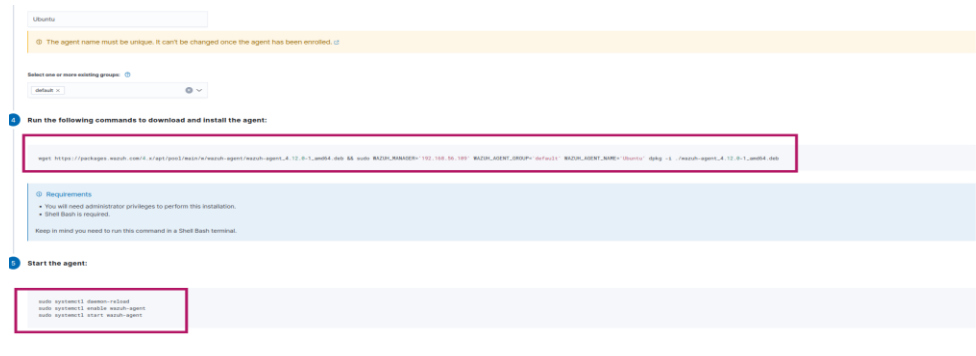


B. Setting Up Wazuh Agent in Another Ubuntu Server

- Click on Deploy Agent from Wazuh Dashboard



- Fill in the required details like: File type, Server IP, Agent Name, Type and then Copy and paste the script that gets generated in Ubuntu Server.



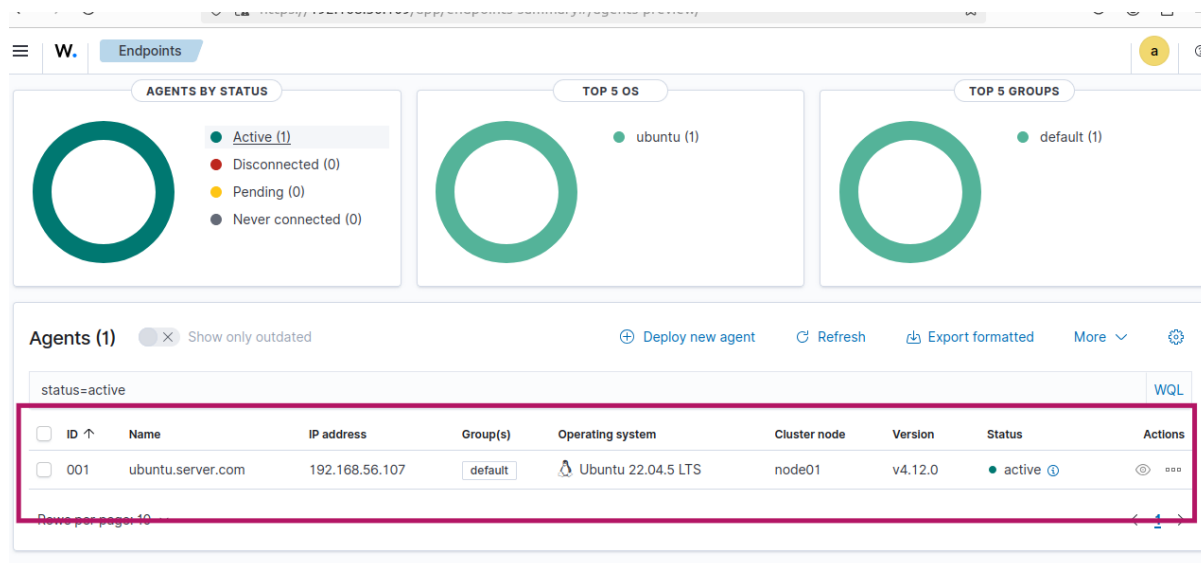
- Once the installation is done run below commands in Ubuntu Server

```
>> sudo systemctl enable wazuh-agent
```

```
>> sudo systemctl start wazuh-agent
```

```
>> sudo systemctl status wazuh-agent
```

- Then you will see agent connected in the Wazuh Dashboard

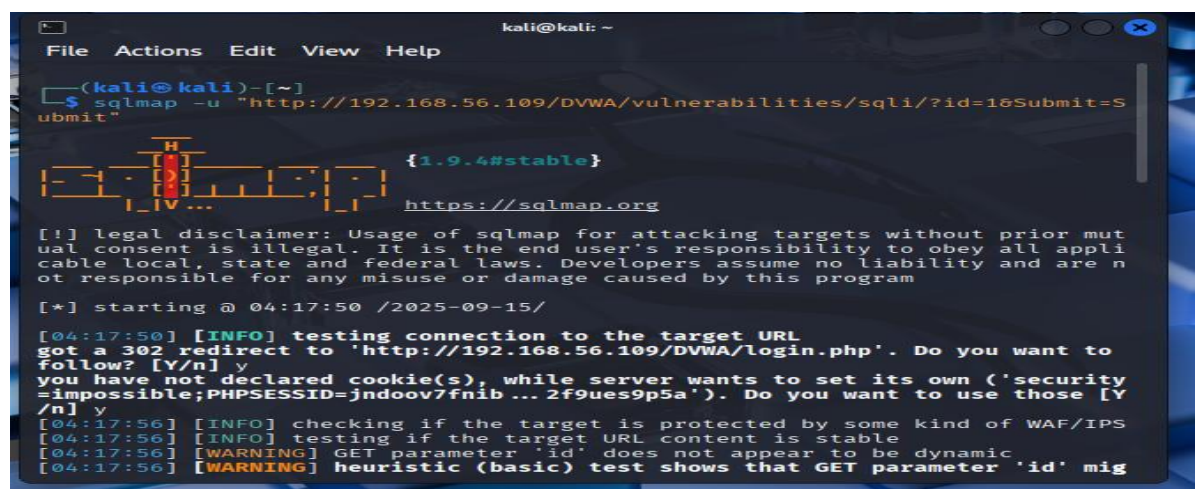


Attack Scenarios & Findings

4.1 SQL Injection ('OR'1'='1--)

Attack:

- Executed manual SQLi via web forms on DVWA and automated attacks using Kali tools (sqlmap).
- WAF (ModSecurity) blocked suspicious payloads; alerts appeared in Wazuh Dashboard.



Screenshot of SQL Attack

96 hits				
Sep 15, 2025 @ 04:03:54.777 - Sep 15, 2025 @ 04:18:54.777				
Export Formatted 643 available fields Columns Density 1 fields sorted Full screen				
timestamp	agent.name	rule.description	rule.level	rule.id
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	SQL injection attempt.	7	31103
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	SQL injection attempt.	7	31103
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	SQL injection attempt.	7	31103
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	SQL injection attempt.	7	31103
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	SQL injection attempt.	7	31103
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	Multiple SQL injection attempts from same source ip.	10	31152
Sep 15, 2025 @ 04:18:11.7...	UbuntuServer	SQL injection attempt.	7	31103

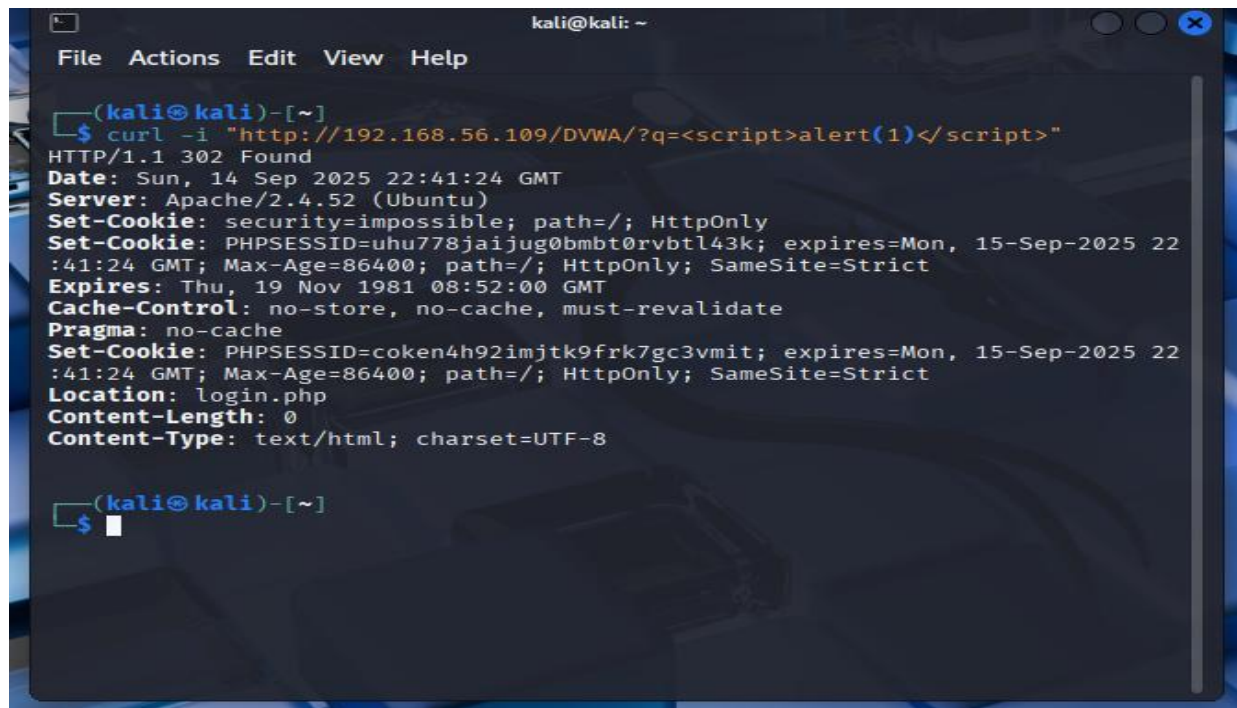
Screenshot of SQL Attack Detection on Wazuh

Result: Successful detection, effective blocking, detailed log message.

4.2 Cross-Site Scripting (XSS;)

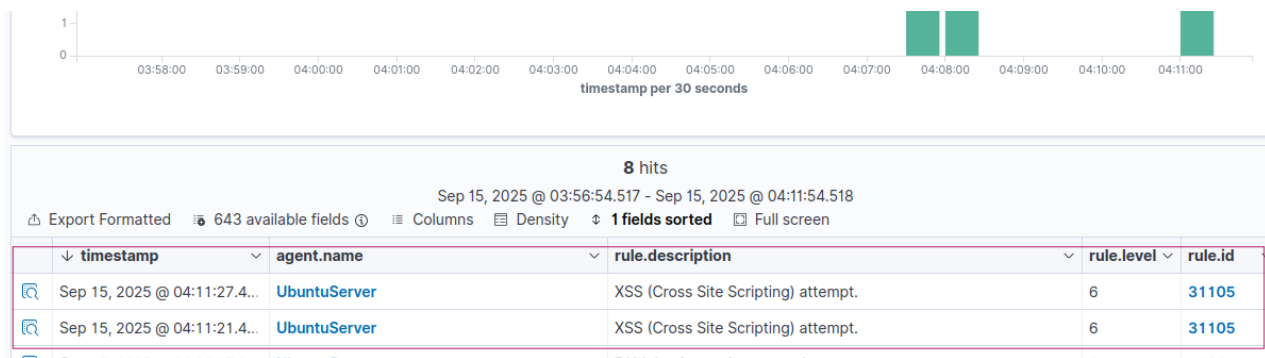
Attack:

- Injected script payload into DVWA fields, monitored for execution and log correlation.



```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ curl -i "http://192.168.56.109/DVWA/?q=<script>alert(1)</script>"  
HTTP/1.1 302 Found  
Date: Sun, 14 Sep 2025 22:41:24 GMT  
Server: Apache/2.4.52 (Ubuntu)  
Set-Cookie: security=impossible; path=/; HttpOnly  
Set-Cookie: PHPSESSID=uhu778jaijug0bmbt0rvbtl43k; expires=Mon, 15-Sep-2025 22:41:24 GMT; Max-Age=86400; path=/; HttpOnly; SameSite=Strict  
Expires: Thu, 19 Nov 1981 08:52:00 GMT  
Cache-Control: no-store, no-cache, must-revalidate  
Pragma: no-cache  
Set-Cookie: PHPSESSID=coken4h92imjtk9frk7gc3vmit; expires=Mon, 15-Sep-2025 22:41:24 GMT; Max-Age=86400; path=/; HttpOnly; SameSite=Strict  
Location: login.php  
Content-Length: 0  
Content-Type: text/html; charset=UTF-8  
  
(kali@kali)-[~]  
$
```

Screenshot of Attack done by Kali Attacker Machine



Screenshot of Attack logs being shown in the Wazuh Dashboard

Result: Alert generated, ModSecurity intercepted attack, visible in Wazuh.

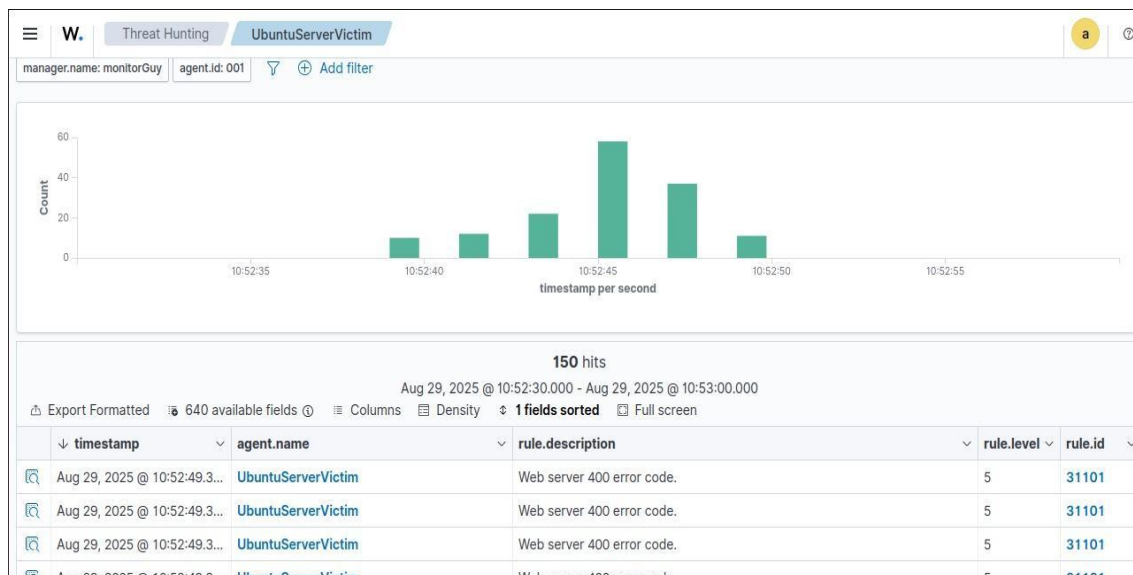
4.3 Brute Force and DoS

Attack:

- Repeated login attempts, rapid request bursts from Kali.

```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ nmap -p80 --script http-brute 192.168.56.10  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-29 01:05 EDT  
Nmap scan report for 192.168.56.10  
Host is up (0.033s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
| http-brute:  
|_ Path "/" does not require authentication  
MAC Address: 08:00:27:DB:20:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 6.16 seconds  
  
(kali@kali)-[~]  
$ nmap -p80 --script http-form-brute --script-args http-form-brute.path=/DVWA/login.php,http-form-brute.method=post,http-form-brute.username=admin,http-form-brute.us  
ervar=username,http-form-brute.passvar=password,http-form-brute.failed="Login failed" 192.168.56.10  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-29 01:13 EDT  
Nmap scan report for 192.168.56.10  
Host is up (0.0016s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
MAC Address: 08:00:27:DB:20:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.10 seconds  
  
(kali@kali)-[~]  
$
```

- Monitored for WAF rate limiting and Wazuh alerting.



Evidence & Result:

- Alerts visible in Wazuh dashboard.

Monitoring & Response

5.1 Wazuh Dashboard

- Showed attack events with source, timestamps, and type (screenshot evidence).
- Allowed investigation, event filtering, and incident verification.

5.2 ModSecurity Logs

- Presented detailed HTTP request logs for blocked threats.

5.3 SOC Workflow

- Attacks simulated from Kali.
- Events visualized and correlated in real time.
- Remediation steps suggested based on findings.

Conclusion

The project demonstrated the effectiveness of combining ModSecurity (as a WAF) and Wazuh (as a SIEM/XDR platform) in detecting, blocking, and monitoring common web-based attacks. The integration provided both prevention at the server level and centralized visibility at the SOC level.

Key takeaways include:

- Layered security matters — ModSecurity blocks attacks in real-time, while Wazuh provides monitoring, alerting, and log correlation.
- Proper configuration is critical — CRS must be included correctly, Wazuh requires sufficient system resources, and logging paths must be validated.
- Resource constraints are real — the Wazuh Indexer can trigger out-of-memory errors on small VMs; allocating ≥ 4 GB RAM or tuning JVM settings is necessary.
- Reproducibility requires documentation — recording exact versions, network settings, and commands ensures others can replicate the lab.

Overall, this lab provides a strong foundation for learning SOC workflows, demonstrating how open-source tools can effectively simulate enterprise-grade detection and response. Future improvements include adding custom correlation rules, automating lab setup with Ansible/Vagrant, and integrating additional log sources (e.g., system logs, IDS alerts) into Wazuh for a more complete SOC simulation.
